

Hydrocarbons

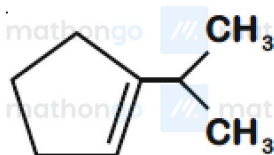
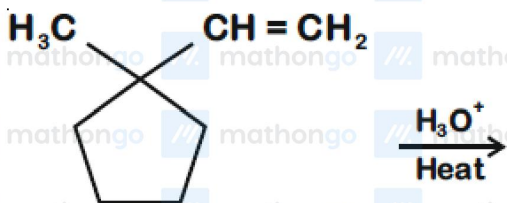
Questions with Answer Keys

JEE Main 2020 Chapterwise

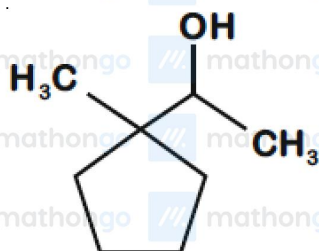
Chemistry

Q1 JEE Main 2020 - 2 September (Morning)

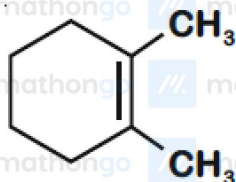
The major product in the following reaction is



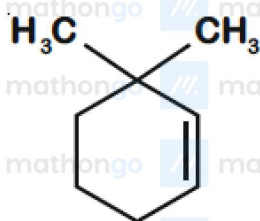
(A)



(B)



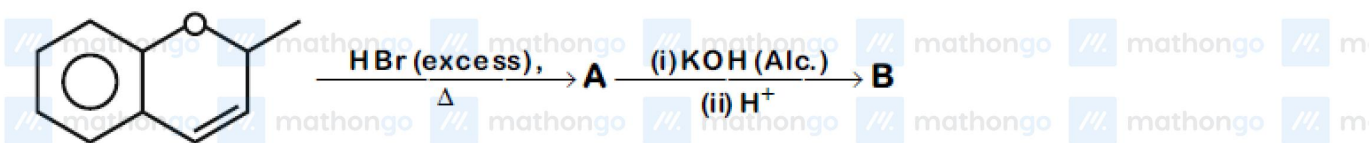
(C)



(D)

Q2 JEE Main 2020 - 2 September (Morning)

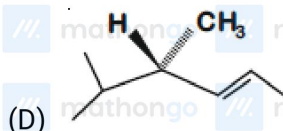
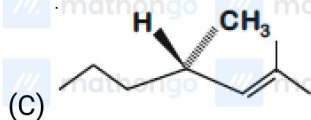
The major aromatic product C in the following reaction sequence will be



Q3 JEE Main 2020 - 3 September (Morning)

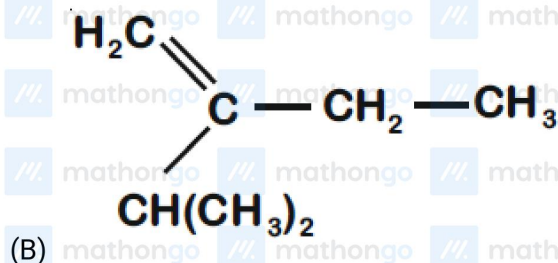
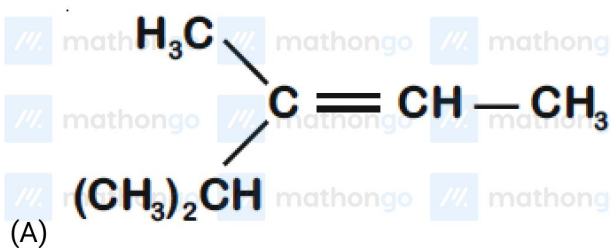
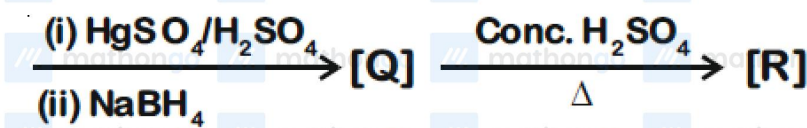
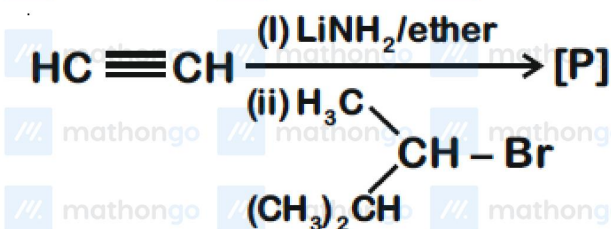
Which of the following compounds produces an optically inactive compound on hydrogenation?





Q4 JEE Main 2020 - 4 September (Evening)

The major product [R] in the following sequence of reactions as:





Q5 JEE Main 2020 - 5 September (Evening)

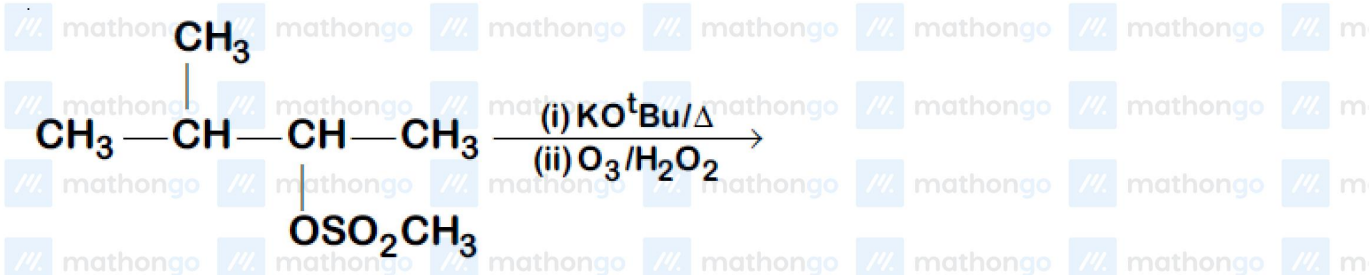
The major product formed in the following reaction is

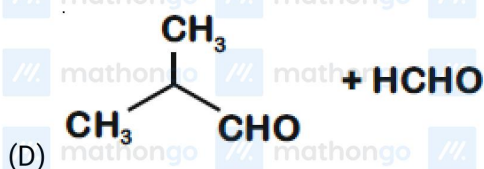
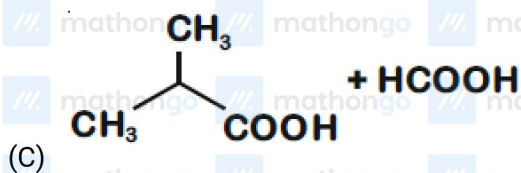
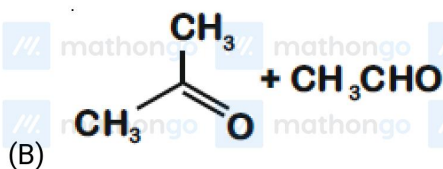
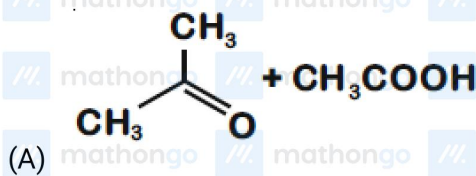


- (A) $\text{Br}(\text{CH}_2)_3\text{CH}(\text{CH}_3)_2$
 (B) $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CH}(\text{CH}_3)_2$
 (C) $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}(\text{CH}_3)_2$
 (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}(\text{Br})(\text{CH}_3)_2$

Q6 JEE Main 2020 - 6 September (Morning)

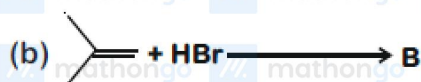
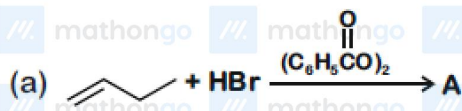
The major products of the following reaction are





Q7 JEE Main 2020 - 6 September (Evening)

The increasing order of the boiling points of the major products *A*, *B* and *C* of the following reactions will be



(A) $A < C < B$

(B) $C < A < B$

(C) $B < C < A$

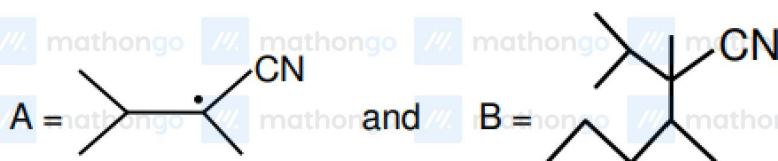
(D) $A < B < C$

Q8 JEE Main 2020 - 8 January (Morning)

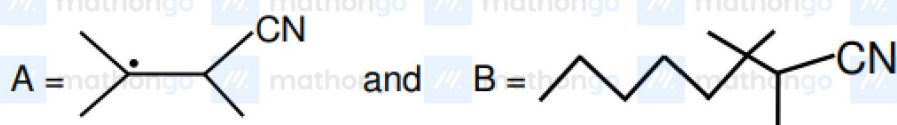
The major products *A* and *B* in the following reactions are :



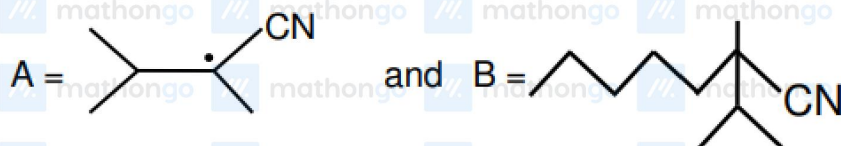
(A)



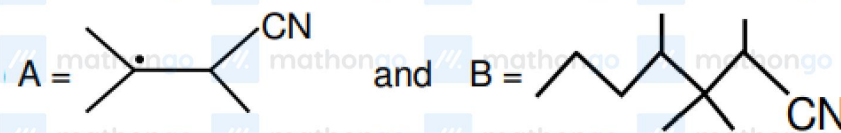
(B)



(C)



(D)



Q9 JEE Main 2020 - 8 January (Evening)

An unsaturated hydrocarbon *X* absorbs two hydrogen molecules on catalytic hydrogenation, and also gives following reaction :

Hydrocarbons

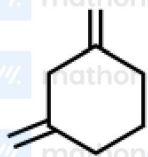
JEE Main 2020 Chapterwise

Questions with Answer Keys

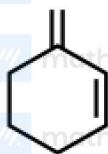
Chemistry

$X \xrightarrow{O_3} A \xrightarrow{[Ag(NH_3)_2]^+} B$ (3-oxo-hexanedicarboxylic acid) X will be :

(A)



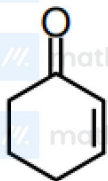
(B)



(C)

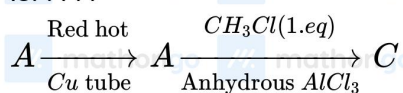


(D)



Q10 JEE Main 2020 - 8 January (Evening)

In the following sequence of reactions the maximum number of atoms present in molecule 'C' in one plane is. . . .



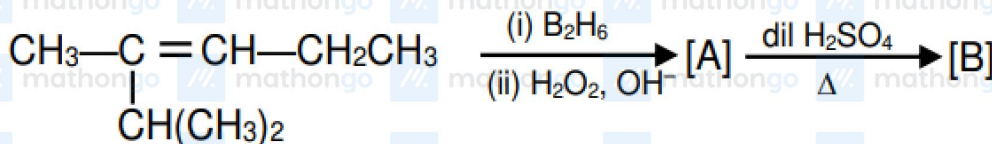
(A is a lowest molecular weight alkyne)

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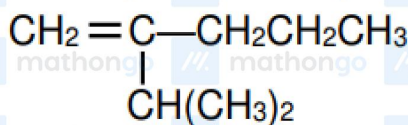
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Q11 JEE Main 2020 - 8 January (Evening)

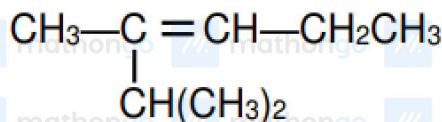
The major product [B] in the following sequence of reactions is :



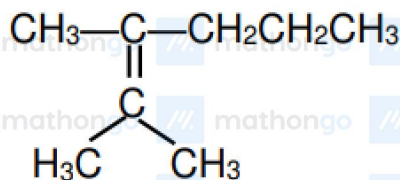
(A)



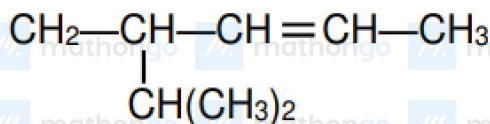
(B)



(C)

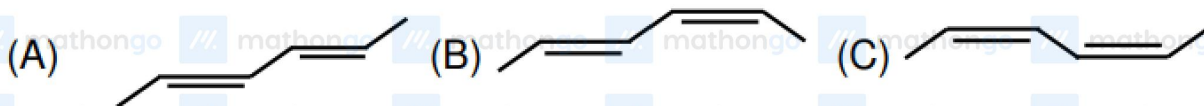


(D)



Q12 JEE Main 2020 - 9 January (Morning)

The correct order of heat of combustion for following alkadienes is :



(A) $C > B > A$

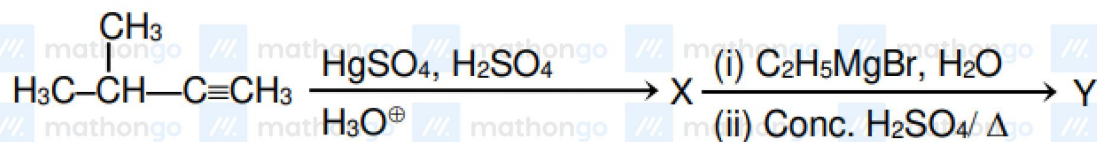
(B) $A > B > C$

(C) $B > A > C$

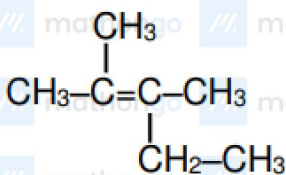
(D) $C > A > B$

Q13 JEE Main 2020 - 9 January (Morning)

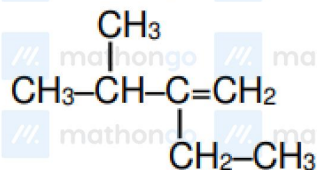
The major product (Y) in the following reactions is :



(A)



(B)

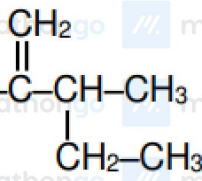


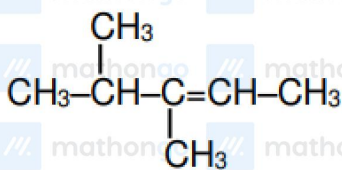
Hydrocarbons

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

(C) 

(D) 

Q14 JEE Main 2020 - 9 January (Evening)

(i) Br_2 , $h\nu$

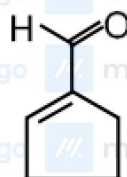
(ii) $\text{KOH}(\text{alc.})$

(iii) O_3

(iv) $(\text{CH}_3)_2\text{S}$

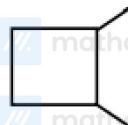
(v) $\text{NaOH}(\text{aq}) + \Delta$

A



In the following reaction A is :

(A) 

(B) 

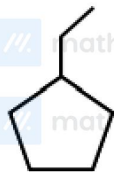
Hydrocarbons

Questions with Answer Keys

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Chemistry

(C) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo



(D) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo



Hydrocarbons

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

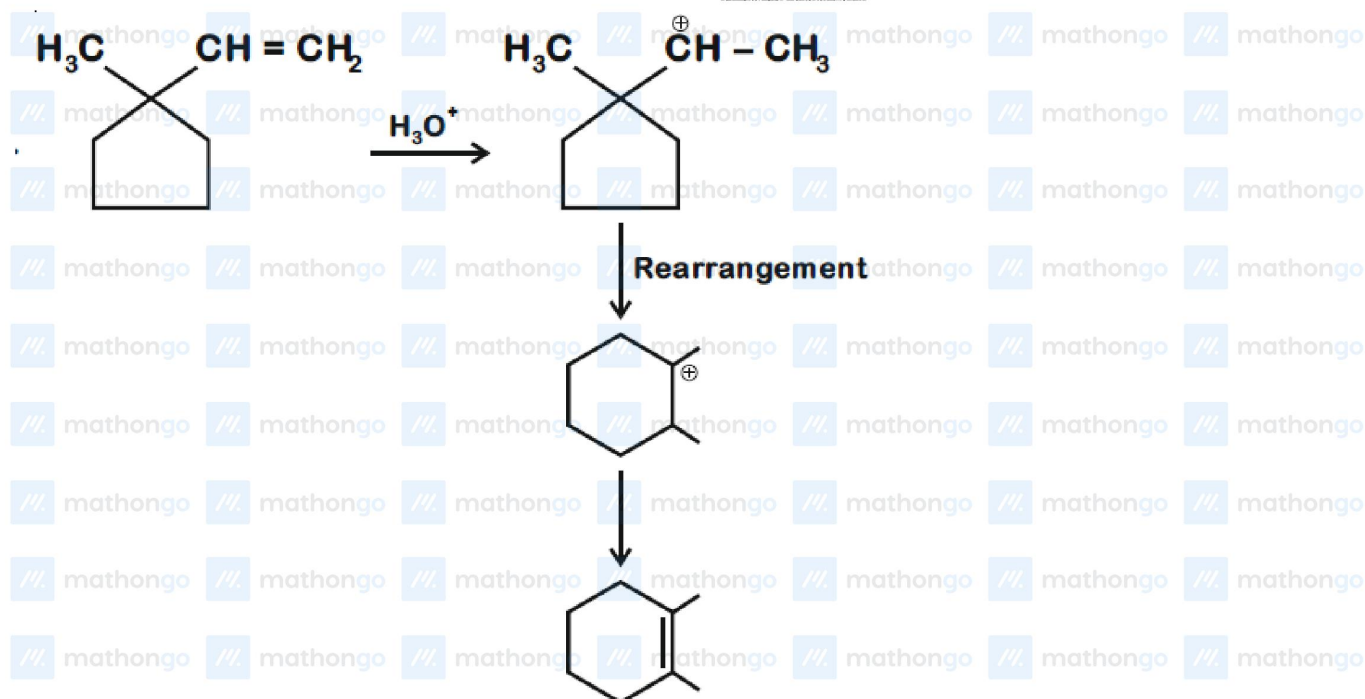
Answer Key

Q1 (C)	Q2 (A)	Q3 (A)	Q4 (C)
Q5 (D)	Q6 (C)	Q7 (C)	Q8 (C)
Q9 (C)	Q10 (13)	Q11 (C)	Q12 (A)
Q13 (A)	Q14 (A)		

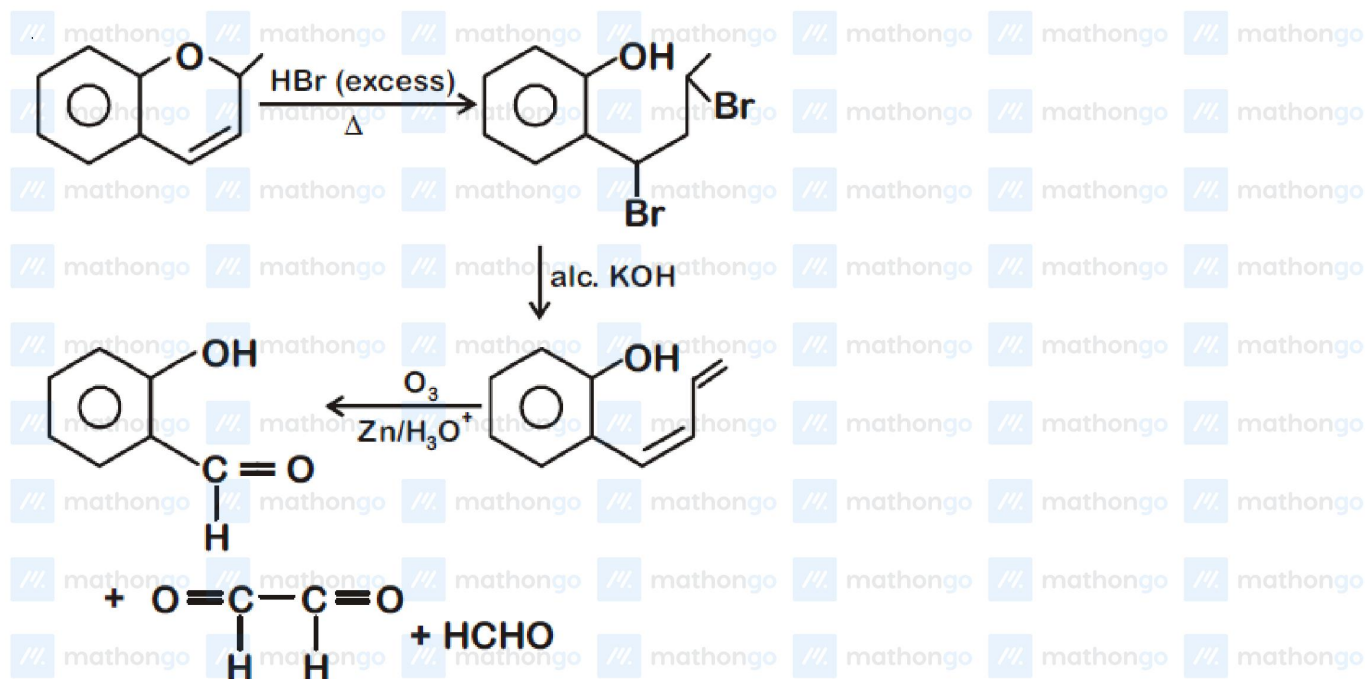
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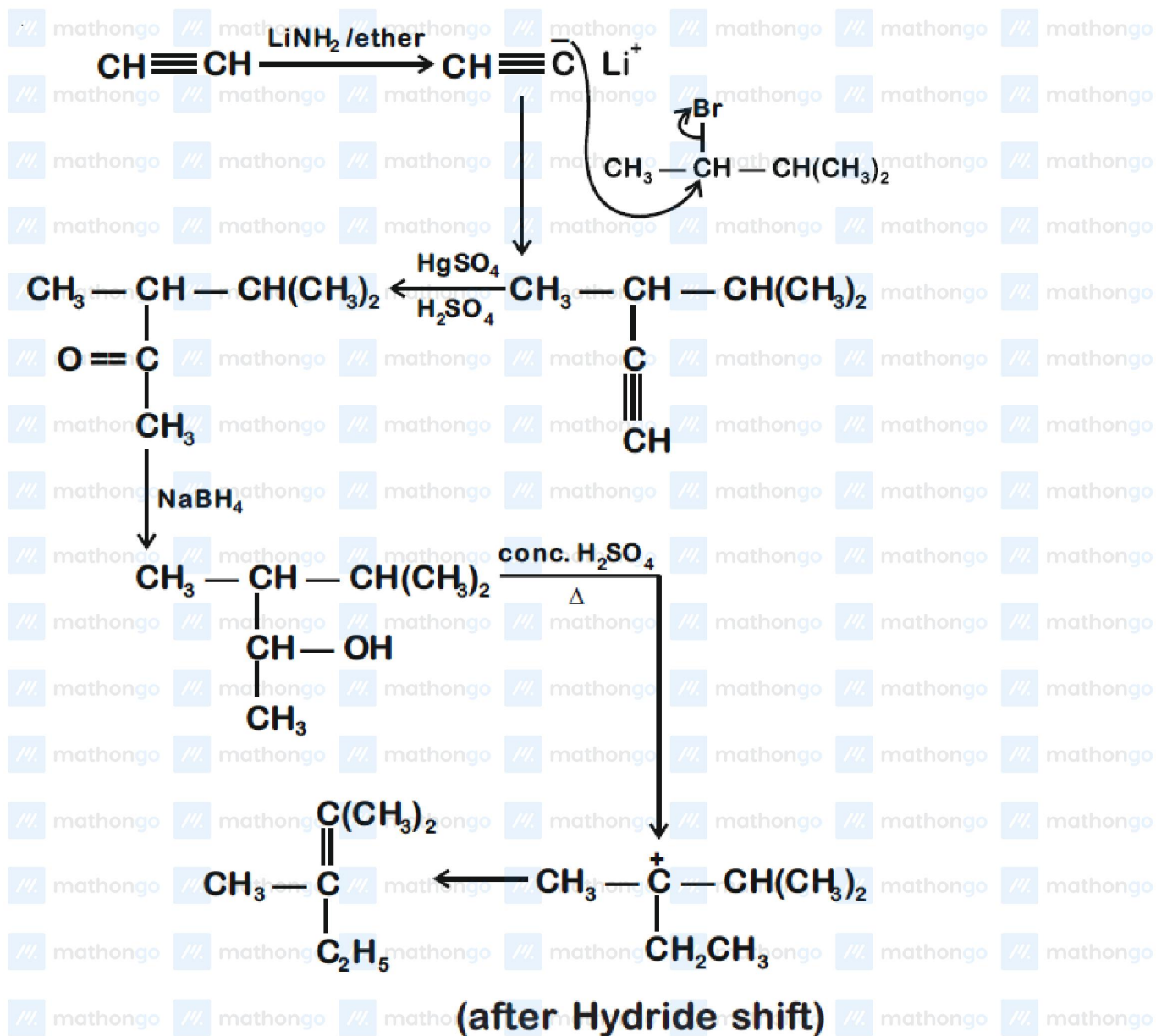
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Q1

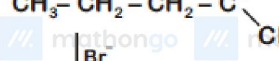
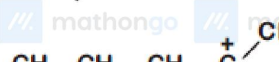
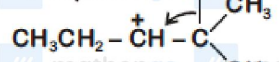
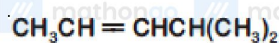


Q2

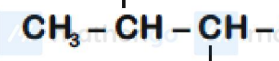
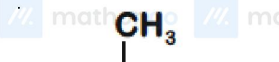




Q5



Q6



Q7

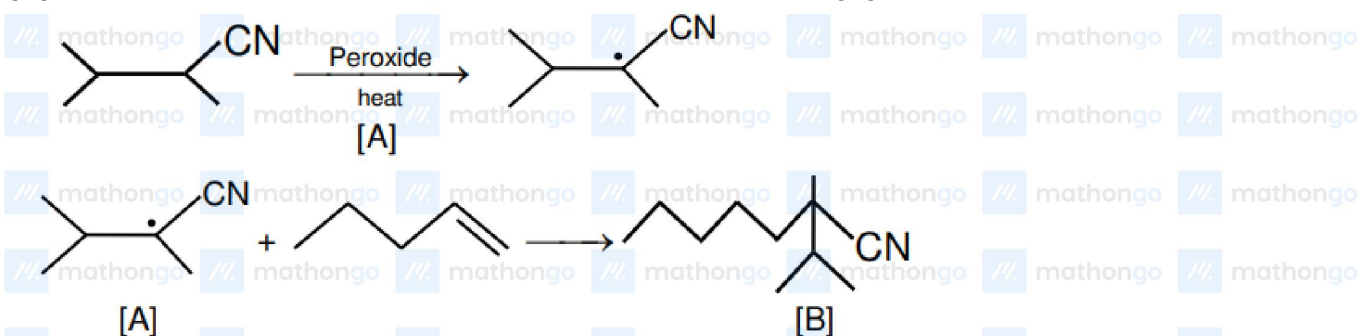


Boiling point decreases with branching

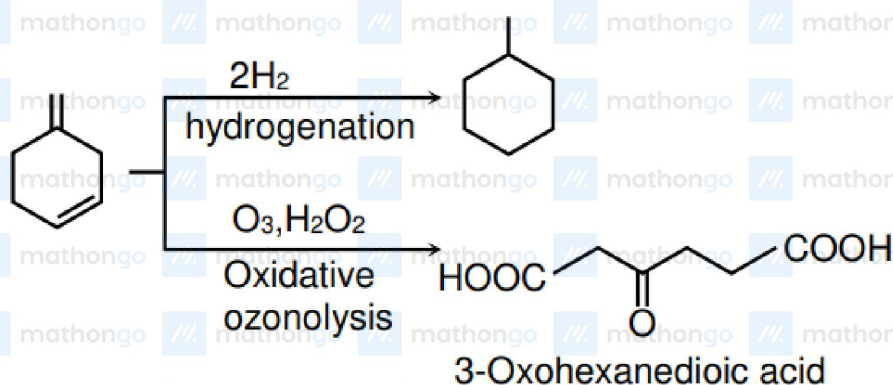
$\therefore$ order of B.P. is $A > C > B$

Q8

[A] is more stable radical and undergoes Markovnikov addition to form [B].

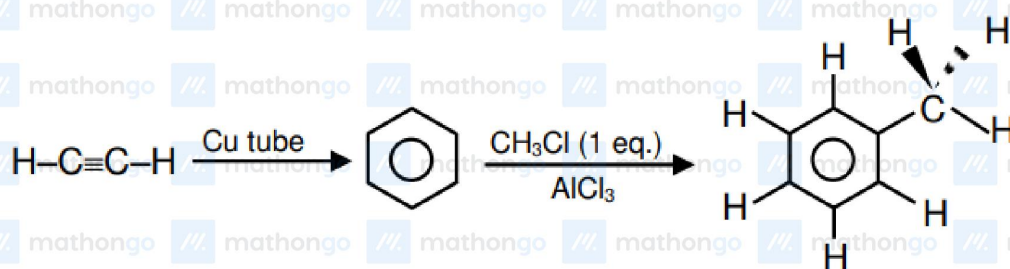


Q9

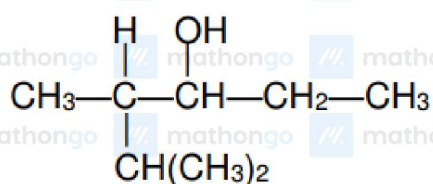
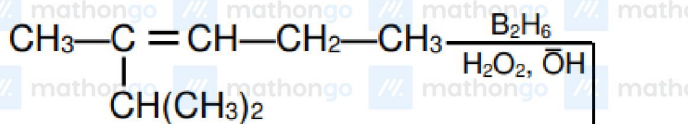


Q10

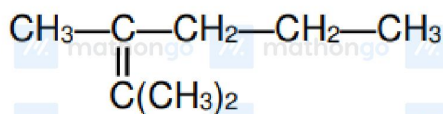
Number of atoms in one plane = 13



Q11



dil $\text{H}_2\text{SO}_4, \Delta$



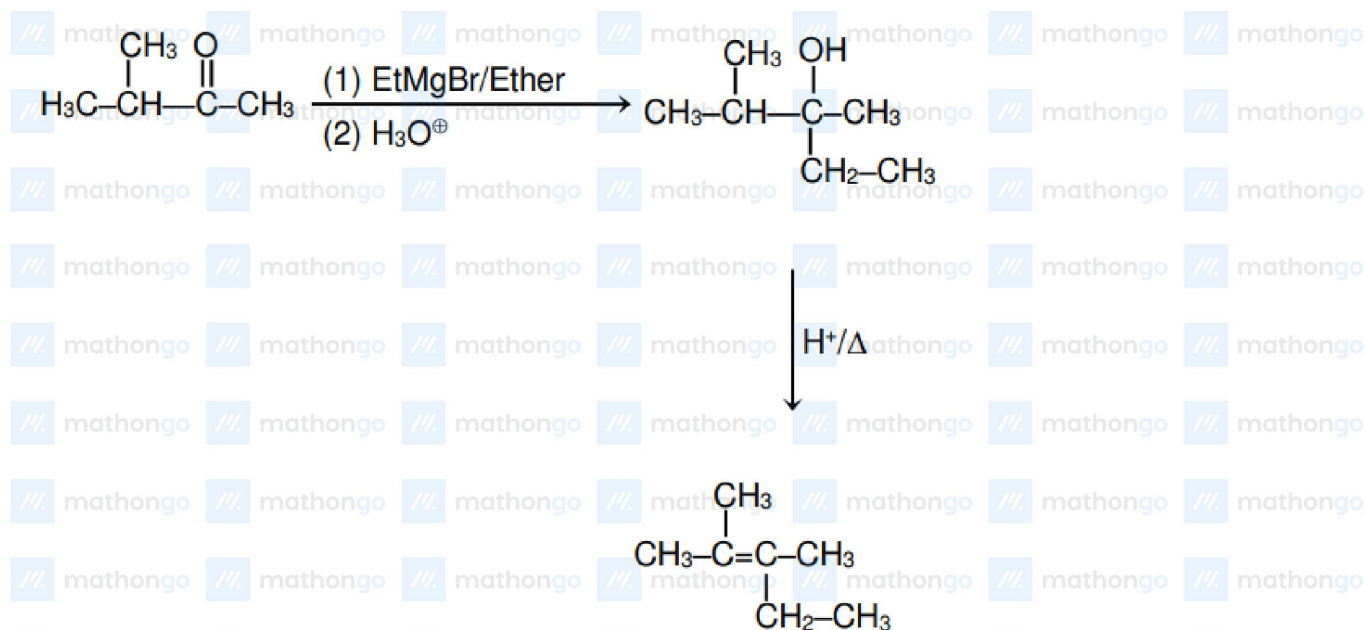
Q12

In isomers of hydrocarbon heat of combustion depends upon their stabilities. As the stability increases heat of combustion decreases.

Stability order



Q13



Q14

